



United International University

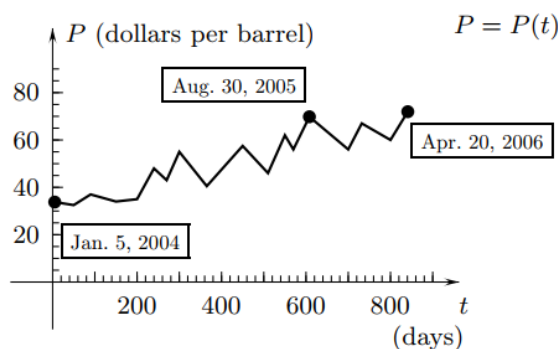
School of Science and Engineering

Sample Questions for Final Exam, Fall-2023

Course Code : Math 1151, Course Title: Fundamental Calculus

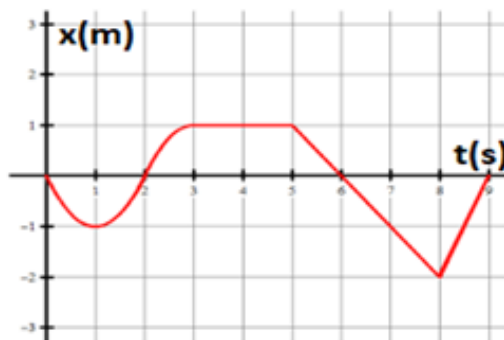
1. (Figure 1) shows the graph of the price $P = P(t)$ (dollars per barrel) of crude oil on the New York Mercantile Exchange with t measured in days and $t = 1$ on January 1, 2004. The price was 33.70 on January 5, 2004 ($t = 5$); 68.81 on August 30, 2005 ($t = 608$); and 71.95 on April 20, 2006 ($t = 841$). What is the average rate of change of the price with respect to time
 - (a) from January 5, 2004 to August 30, 2005
 - (b) from August 30, 2005 to April 20, 2006

Figure 1:



2. The following figure (Figure 2) represents a position function of a particle at time t .

Figure 2:



- (a) Find the average velocities over the time intervals $[1, 3]$ and $[5, 8]$.
- (b) Find the value(s) of t at which the instantaneous velocity is zero.
- (c) On what interval(s), is the position graph increasing, decreasing, and stand-still?
- (d) Roughly sketch the velocity graph of the particle.

3. The graphs in Figure 3 of the functions $g(x)$ is given.

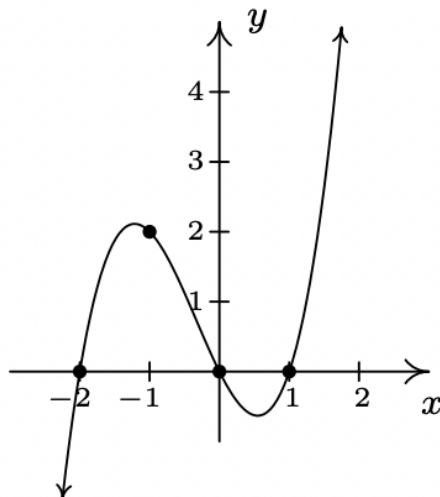


Figure 3:

- (a) State the sign of $g'(x)$ at the specified points.
 - (b) Arrange them in an increasing sequence.
 - (c) Find the values of x at which $g'(x) = 0$. Explain the reason.
 - (d) Sketch the graph of $g'(x)$.
4. The position function of a moving particle is given by $s(t) = t^2 - 2$, where s is in meters and t is in seconds.
- (a) Find the average velocity of the particle over the time interval $[2, 3]$.
 - (b) Find the particle's instantaneous velocity at time $t = 2$ s.
 - (c) Draw the position function $s(t)$, and velocity function $v(t)$ in the same plot.
5. Consider the following functions:

(a) $f(x) = x^3 + 3$

(b) $g(x) = 6x - x^2$

- i. Find the instantaneous rate of change of $f(x)$ and $g(x)$ with respect to x at an arbitrary value of x .
- ii. Use (i) to find the slope of the tangent lines for $x = 1$.
- iii. Find the equation of the tangent lines to the graphs at $x = 1$.
- iv. Find the equation of the secant lines in the interval $[-1, 1]$.
- v. Draw the graph of $f(x)$ and $g(x)$ together with the tangent lines and secant lines.

6. Consider the following functions:

(a) $f(x) = -3x^2$

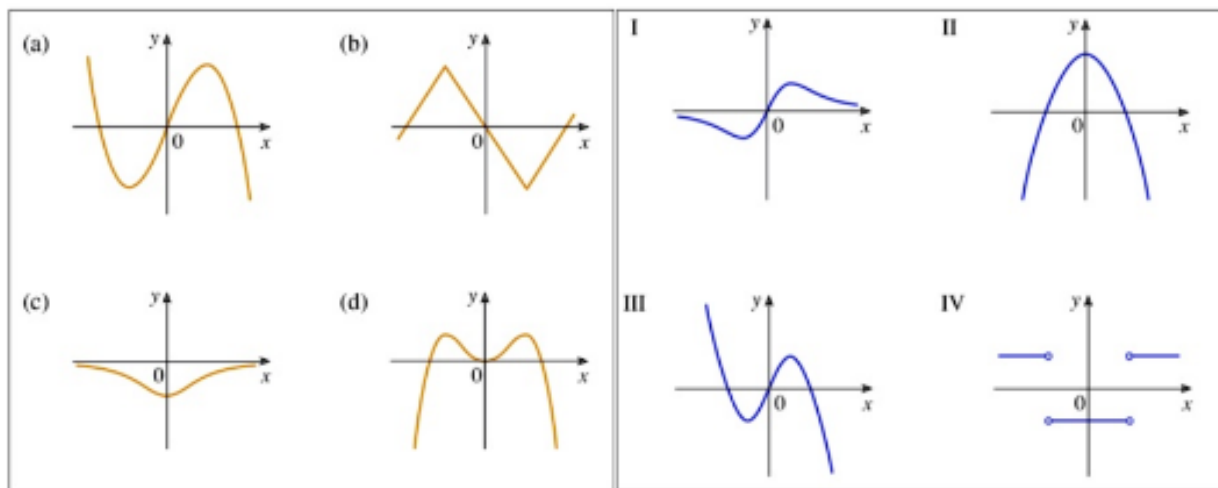
(b) $g(x) = -4 + x^3$

- i. Find the derivative of the functions with respect to x by using the formula

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

- ii. Find the equation of the tangent lines to the functions at $x = -1$.
 - iii. Draw the functions and the tangents on the same plot.
7. Match the graphs of the functions shown in (a–d) (Figure 4) with the graphs of their derivatives in (I–IV) (Figure 4).

Figure 4:



8. Sketch the graph of the derivative of each function in Figure 5.

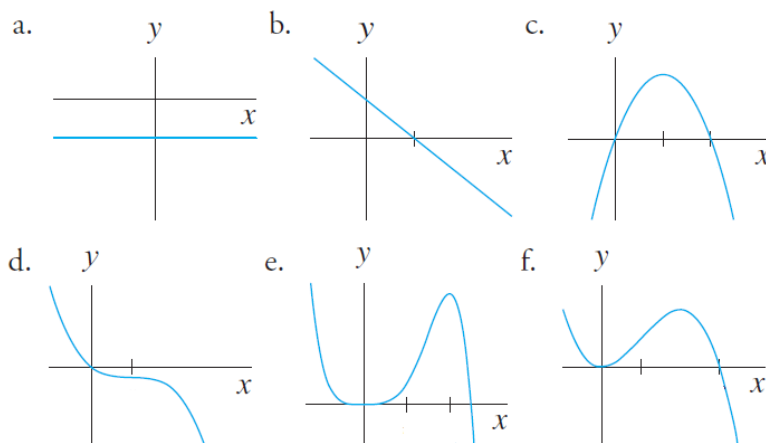


Figure 5:

9. Consider the function

$$f(x) = \begin{cases} 4 - x & \text{if } x < 0 \\ 4 - x^2 & \text{if } x \geq 0 \end{cases}$$

- (a) Determine whether the function $f(x)$ is continuous and differentiable at $x = 0$.
 - (b) Sketch the graph of $f(x)$.
 - (c) Sketch the graph of $f'(x)$.
10. The equation of motion of a particle is $s = t^4 - 2t^3 + t^2 - t$, where s is in meters and t is in seconds.

- (a) Find the velocity and acceleration as functions of t .
- (b) Find the acceleration after 1 s.
11. The graph of $f(x)$ is given in Figure 6. State, with reasons, the numbers at which $f(x)$ is not differentiable.

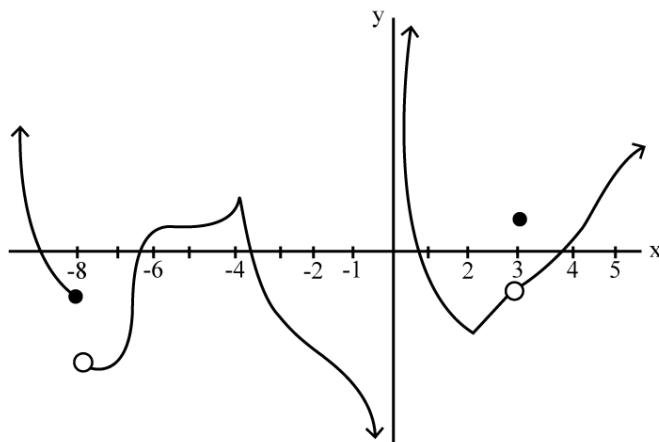


Figure 6:

12. The Figure 7 shows the graphs of f , f' , and f'' . Identify each graph with proper reasoning.

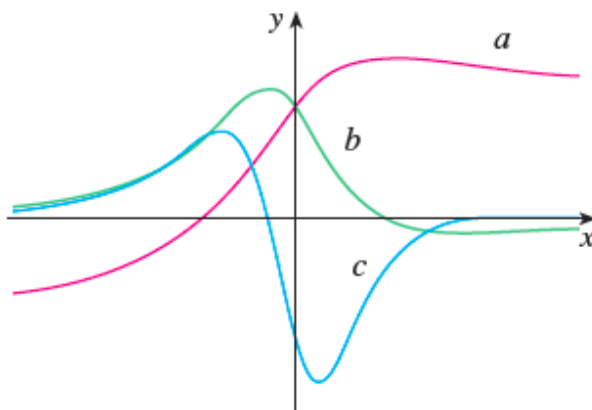


Figure 7:

13. (a) At what point(s) on the curve $y = x^3 - 3x^2 + 7$ is the tangent line parallel to the line $y = 9x - 1$?
- (b) Find all points on the graph of the function $f(x) = x^3 + 3x^2 - 9x + 10$ at which the tangent line is horizontal.
- (c) At what point on the curve $y = \sqrt{1 + 2x}$ is the tangent line perpendicular to the line $6x + 2y = 1$?
14. Suppose $f(3) = 7$, $f'(3) = 6$, $g(3) = -5$, and $g'(3) = -2$. Find $h'(3)$ for the followings.
- (a) $h(x) = 3f(x) - 2g(x) + f(x)g(x)$ (c) $h(x) = e^x g(x)$
- (b) $h(x) = \frac{f(x)}{\ln x}$ (d) $h(x) = \frac{1 + \sqrt{x}f(x)}{x^2 + g(x)}$

15. (a) Consider $f(x) = \sec^3(x^5) - x^2 \cos 3x - 5$. find $f'(x)$.
 (b) If $t = \ln(\sin(y^2 + 1))$ and $y = \cos(x^2 e^x)$, find $\frac{dt}{dx}$.
16. If $xg(x) + \pi \cos^2 2x = k$, where k is a constant, and $g(\frac{\pi}{2}) = 1$, find $g'(\frac{\pi}{2})$.
17. Write the composite function in the form of $f(g(x))$. [Identify the inner function $u = g(x)$ and the outer function $y = f(u)$]. Then, find $\frac{dy}{dx}$.
- (a) $y = \ln(\cos x)$ (b) $y = \sqrt[3]{e^{2x} + 1}$ (c) $y = (x^5 + 3x^2 + 1)^{-10}$
18. Find an equation for the tangent line to the graph of $y = \cos(\cos x)$ at the point $x = \frac{\pi}{2}$.
19. Differentiate the following functions:
- (a) $f(x) = \sec^3(\frac{x}{\sqrt{1-x^2}})$ (b) $g(x) = (x^7 - 3)^{-2} \tan(\frac{3}{x})$
20. The following table gives some information about $f(x)$, $f'(x)$, $g(x)$ and $g'(x)$ at x .

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-3	4	7	-2	-1
0	7	-1	3	-2
2	-1	2	-3	5

- If $u(x) = f(x)g(x)$, $v(x) = \frac{f(x)}{g(x)}$, $w(x) = f(g(x))$, and $z(x) = g(x^3 - 2x - 2)$, then use the Chain Rule to evaluate $u'(-3)$, $v'(0)$, $w'(2)$, and $z'(2)$.
21. Graph the function over the specified interval. Then use simple area formula from geometry to find the area function $A(x)$ that gives the area between the graph of the specified function $f(x)$ and the interval $[a, x]$. Confirm that $A'(x) = f(x)$. Also, verify your answer by direct integration.
- (a) $f(x) = 6 - x$; $[a, x] = [-2, x]$
 (b) $f(x) = 10$; $[a, x] = [2, x]$
 (c) $f(x) = 2x + 4$; $[a, x] = [\frac{1}{2}, x]$
22. Use the anti-derivative method to find the area under the graph of $y = x^3 - 1$ over the interval $[0, 2]$.
23. Consider the function:

$$f(x) = \begin{cases} 2x + 3 & \text{if } x \leq 0 \\ 3 & \text{if } x > 0 \end{cases}$$

Use $f(x)$ to evaluate: $\int_{-3/2}^2 f(x) dx$

24. The graph of $f'(x)$ is given in Figure 8. Sketch the graph of the continuous function $f(x)$ on $[0, 6]$; where $f(0) = -2$.

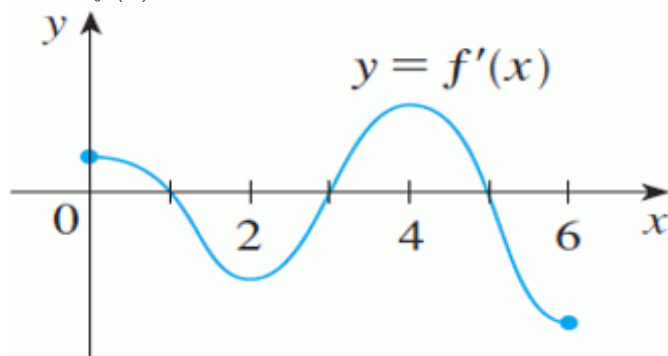


Figure 8:

25. Sketch the region whose signed area is represented by the definite integral, and evaluate the integral using an appropriate formula from geometry, where needed:

(a) $\int_0^2 |3x - 6| dx$

(c) $\int_{-3}^3 \sqrt{9 - x^2} dx$

(b) $\int_{-5}^0 |2x + 5| dx$

(d) $\int_0^4 -\sqrt{(16 - x^2)} dx$

26. Evaluate $\int_0^3 (2x + \sqrt{9 - x^2}) dx$

27. Consider the function:

$$f(x) = \begin{cases} |x + 4| & \text{if } x \leq 0 \\ -x + 4 & \text{if } x > 0 \end{cases}$$

Use $f(x)$ evaluate the following integrals:

(a) $\int_{-8}^{-5} f(x) dx$

(b) $\int_{-4}^4 f(x) dx$

(c) $\int_4^6 f(x) dx$

28. The graph of f is shown in Figure 9. Evaluate the following integrals.

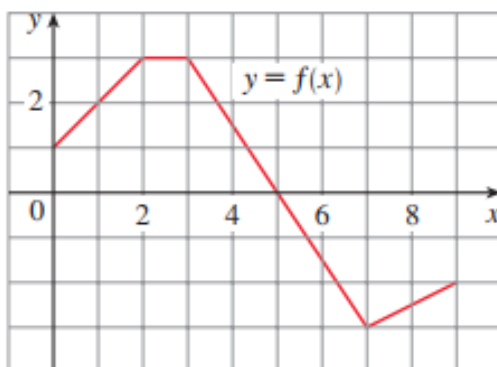


Figure 9:

(a) $\int_0^3 f(x) dx$

(b) $\int_5^8 f(x) dx$

(c) $\int_3^7 |f(x)| dx$

29. If $\int_2^8 f(x) dx = 7.3$ and $\int_2^4 f(x) dx = 5.9$, find $\int_4^8 f(x) dx$.

30. Evaluate the following integrals:

- | | | |
|--|---|---|
| (a) $\int_{-1}^1 t(1-t)^2 dt$ | (i) $\int \frac{e^x}{3+e^{2x}} dx$ | (q) $\int \frac{x}{\sqrt{4-5x^2}} dx$ |
| (b) $\int_1^4 (\frac{4+6u}{\sqrt{u}}) du$ | (j) $\int \frac{1}{4+25x^2} dx$ | (r) $\int_0^4 3^s ds$ |
| (c) $\int x^3 \sqrt{2+x^4} dx$ | (k) $\int \frac{x^4}{\sqrt{2+x^5}} dx$ | (s) $\int \frac{e^x+e^{-x}}{e^x-e^{-x}} dx$ |
| (d) $\int x\sqrt{x+1} dx$ | (l) $\int \frac{\cos(\frac{5}{x})}{3x^2} dx$ | (t) $\int \frac{\sin x}{\cos^2 x+1} dx$ |
| (e) $\int x^{\frac{1}{3}}(1+x^2) dx$ | (m) $\int \frac{e^{-\sqrt{2x}}}{2\sqrt{2x}} dx$ | (u) $\int \sin(\sin x) \cos x dx$ |
| (f) $\int \frac{x^3+2x-5}{x^2} dx$ | (n) $\int \cos^4 3x dx$ | (v) $\int \sin^3 x \cos x dx$ |
| (g) $\int \frac{\sin x}{\cos^2 x} dx$ | (o) $\int_{-\sqrt{2}}^{-2/\sqrt{3}} \frac{1}{x\sqrt{x^2-1}} dx$ | (w) $\int_0^{\frac{3\pi}{2}} \sin x dx$ |
| (h) $\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} dx$ | (p) $\int \frac{x}{3+7x^2} dx$ | (x) $\int_0^{\frac{\pi}{4}} \sec^6 x \tan^6 x dx$ |

31. Find the area of the following couple of curves by:

- | | |
|-------------------------------------|-------------------------------------|
| (a) integrating with respect to x | (b) integrating with respect to y |
| i. $y^2 = 4x$ and $y = x - 8$ | iv. $y = 1 - x^2$ and $y = 0$ |
| ii. $y = 1 + x$ and $y = 2x^2$ | v. $y = 3x$, $y = -3$ and $x = 3$ |
| iii. $y^2 = 2x$ and $x + y = 4$ | vi. $y = 1 - x^2$, and $x + y = 1$ |

32. Questions from 7.2

Evaluate the following integrals:

- | | | |
|--------------------------------------|---|--|
| (a) $\int x^2 \ln x dx$ | (g) $\int \frac{\sqrt{x^2-9}}{x^3} dx$ | (m) $\int \frac{x}{x^2+2x+2} dx$ |
| (b) $\int \frac{xe^x}{(1+x)^2} dx$ | (h) $\int_0^{0.3} \frac{x}{(9-25x^2)^{\frac{3}{2}}} dx$ | (n) $\int_0^4 \sqrt{x(4-x)} dx$ |
| (c) $\int x^2 \cos x dx$ | (i) $\int \frac{1}{x^2\sqrt{x^2+25}} dx$ | (o) $\int_1^{\ln e} xe^{-2x} dx$ |
| (d) $\int x \tan^{-1}(3x) dx$ | (j) $\int \frac{1}{\sqrt{x^2+4x+5}} dx$ | (p) $\int_0^{\pi/3} x \sin x dx$ |
| (e) $\int e^{-2x} \sin 3x dx$ | (k) $\int \frac{\sqrt{x-3}}{x} dx$ | (q) $\int_0^e \ln(x+e) dx$ |
| (f) $\int \frac{\ln x}{\sqrt{x}} dx$ | (l) $\int \frac{1}{2x^2+4x+7} dx$ | (r) $\int \frac{x^5-3x^2+2}{x^3-x} dx$ |

33. Find the area of the shaded region in the following figures (Figure 10):

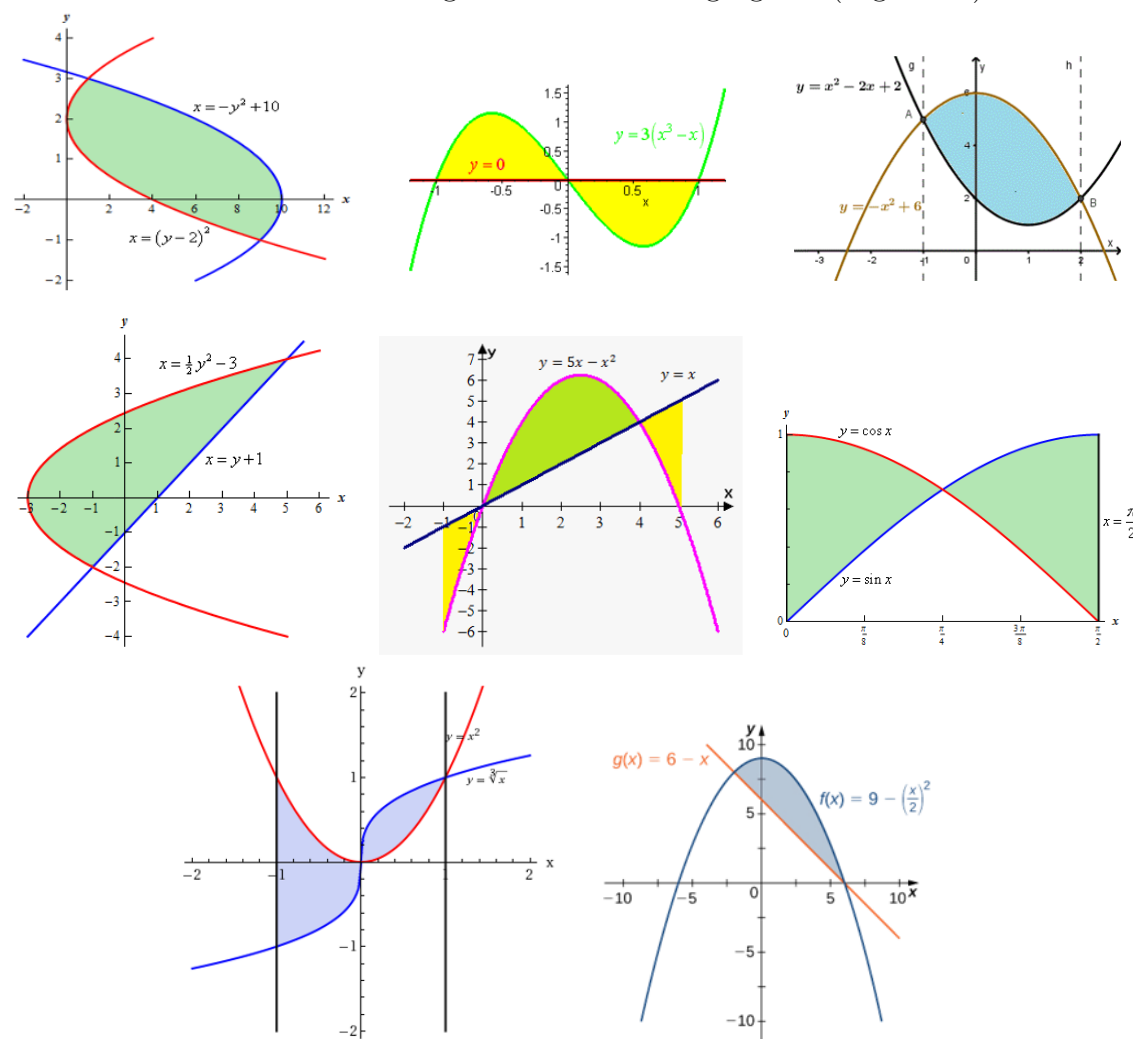


Figure 10: